Can Sovereign Wealth Fund Disclosures Be Monetized?
A Disclosure-Lag-Aware Study of PIF and Norges Bank Signals

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Abstract—This paper examines whether public sovereign wealth fund (SWF) disclosures contain investable information once disclosure lag, partial visibility, and implementation constraints are treated as first-class design variables. The project began with three funds—Saudi Arabia's Public Investment Fund (PIF), Norges Bank Investment Management (NBIM), and Singapore's GIC—but quickly encountered radical disclosure asymmetry: NBIM provides broad but slow public-equity visibility, PIF provides security-level visibility only for its reportable U.S. 13F sleeve, and GIC is too opaque to support a comparable holdings panel. We therefore construct fund-specific normalized datasets and test lag-aware strategies rather than forcing all funds into a common template.

For PIF, five backtests are implemented on a canonical 13F panel with filing-aware trade timing, benchmark comparison against SPY, and a corrected split-adjusted price layer. After validation, four of the five PIF strategies are positive in absolute return, but all five still underperform SPY on an excess-return basis. For NBIM, eight strategies are tested against VT using split-adjusted daily prices for first-post-release execution and month-end marks derived from the same daily source. The strongest direct mirror results are shown to be contaminated by survivor-selection in a bounded mega-cap universe. More credible sector-based tests produce only modest benchmark-relative gains, with the best results coming from industry weight-change and concentrated leadership strategies. A final robustness layer then stresses the surviving signals with execution lags, explicit one-way costs, realistic caps, subperiod splits, and alternate benchmarks. An attribution layer decomposes benchmark-relative performance into exposure timing, allocation, and concentration effects, while a final inference layer adds formal significance tests, bootstrap confidence intervals, turnover-adjusted performance metrics, and an expanded benchmark stack that includes a cash hurdle and defensive-equity comparators. Across both funds, the evidence does not support strong claims of easy copy-trading alpha. The most credible residual signal is slower-moving sector posture rather than single-name imitation. Under the expanded benchmark lens, NBIM's concentrated leadership sleeve remains the strongest return-seeking candidate, while the combined exposure-regime overlay is better interpreted as a lower-beta reference design than as a benchmark-beating alpha engine.

Index Terms—sovereign wealth funds, 13F, NBIM, PIF, disclosure lag, backtesting, sector rotation, alpha

I. Introduction

Large sovereign wealth funds (SWFs) are unusual public investors. They are patient, structurally large, politically salient, and often treated by market participants as "strong hands" rather than tactical traders. This creates an intuitive hypothesis: if SWFs are informed, patient allocators, perhaps public disclosure of their holdings or reallocations contains tradable information.

That simple hypothesis becomes difficult once one confronts the actual disclosure regimes. Some funds disclose broad holdings but at long lag; others disclose only narrow regulatory subsets; others remain too opaque to support a serious panel. This project was designed around that asymmetry rather than around an idealized "all funds disclose the same thing" assumption.

The original target funds were:

- NBIM, manager of Norway's Government Pension Fund Global;
- PIF, Saudi Arabia's Public Investment Fund; and
- GIC, Singapore's sovereign wealth fund.

The original research questions were:

1) Can an investor harvest alpha by mirroring disclosed SWF positions after the information becomes public?
2) If single-name mirroring fails, can alpha still be extracted from the sectors or industries into which SWFs are rotating?
3) Are observable exits or reductions structured enough to improve timing or reduce stale-trade risk?

The project ultimately contributes four things:

1) a disclosure-aware research design for heterogeneous SWF datasets;
2) normalized holdings, transition, and benchmark layers for PIF and NBIM;
3) validated lag-aware backtests distinguishing exploratory from credible results; and
4) reproducible reports, charts, and interactive artifacts that expose the full path from disclosure to simulated trade.

TABLE I
Disclosure Regimes Across Target Funds

Fund	Primary Public Source	Research Implication
NBIM	Public holdings database plus annual / half-year reports	Broad public-equity coverage, but low timeliness; suitable for slow-moving portfolio and industry tests.
PIF	SEC 13F filings	Security-level visibility with filing dates, share counts, and values; strongest candidate for lagged mirroring tests.
GIC	Annual reports and sparse external ownership filings	Too opaque for a comparable holdings panel; retained as a qualitative disclosure-asymmetry case.

II. Research Scope and Disclosure Regimes

A. Core Principle

The governing design rule was:

> A signal may only be traded after it is public, and no backtest may use a portfolio effective date as a trade date unless it is also the public date.

This rule is the main protection against overstated alpha in disclosure-based research.

B. Disclosure Regimes

Table I summarizes the three target funds.

The final quantitative panel therefore excludes GIC, not because the fund is unimportant, but because the public data was too thin to support a comparable holdings backtest without false precision [4], [5].

III. Data Construction Pipeline

A. Data Layers

The project was built around reusable data layers rather than one-off spreadsheets:

- Holdings panels: normalized security or issuer snapshots.
- Transition panels: entries, exits, and change classifications between snapshots.
- Summary panels: sector, geography, and concentration summaries by period.
- Backtest outputs: signal eligibility, rebalance events, order blotters, position histories, portfolio paths, and benchmark-relative summaries.

B. Dataset Scale

The final project scale is summarized in Table II.

C. PIF Data Pipeline

PIF data was pulled from EDGAR 13F filing folders containing primary_doc.xml and informationtable.xml, consistent with the SEC's 13F disclosure framework [1]. The pipeline:

1) parsed filing metadata and holdings rows from XML,
2) indexed amendments and canonical filing versions,
3) normalized security keys using CUSIP, class, and option flags,
4) built a transition dataset based on share-count and value changes, and
5) mapped approved securities to a validated price layer.

TABLE II
Dataset Scale After Cleaning

Fund	Holdings Rows	Periods	Additional Scale Information
PIF	775	30	30 canonical filings, 6 amendments in raw set, 998 transition rows, 88 priced securities.
NBIM	97,010	11	98,450 transition rows, 16,445 unique issuer-country keys, 22 priced instruments with full daily coverage.

The transition dataset classified entry_observed, exit_observed, likely_accumulation, likely_reduction, and continued_holding. Because 13F includes share counts, these classifications could be anchored in actual share changes rather than inferred ownership percentages.

D. NBIM Data Pipeline

NBIM data was assembled from downloaded public equity files labeled by snapshot date (e.g., eq_YYYYMMDD) using NBIM's public holdings distribution [2], [3]. The pipeline:

1) stacked semicolon-delimited holdings files into a consolidated issuer-level panel,
2) normalized numeric ownership and voting fields,
3) generated annual and half-year summary datasets,
4) classified issuer transitions between snapshots, and
5) mapped industry exposures to liquid U.S. sector ETF proxies.

Unlike PIF, NBIM holdings did not include share counts. This matters. A change in market value alone does not prove buying or selling. Therefore, the NBIM transition engine treated inclusion/exclusion and ownership-percentage changes as the strongest available indicators and did not infer flows directly from value movement.

E. GIC Treatment

GIC remained in scope conceptually but out of scope for the quantitative backtests. The project documented the relevant proxy channels—major-shareholder filings, 13D/G, HKEX DI, FCA TR-1, EDINET, and deal press—but did not construct a comparable holdings panel because the public evidence was episodic, threshold-driven, and structurally incomplete.

IV. Price Layers, Benchmarks, and Trade Timing

A. Trade Timing

All strategies obeyed disclosure-lag-aware execution rules:

- PIF signal date: SEC public filing date.
- PIF trade date: next trading day after the public date.
- NBIM signal date: official annual or half-year publication date.
- NBIM trade date: first tradable close strictly after publication.

For NBIM, the portfolio is rebalanced at the first post-release daily close and then carried at month-end closes until the next report. This preserves realistic post-disclosure reaction timing without requiring full daily path simulation between reports.

B. Benchmarks

The benchmark selection was intentionally layered. The primary layer remains the market opportunity-cost comparison:

- PIF benchmark: SPY, reflecting a U.S.-listed 13F sleeve.
- NBIM benchmark: VT, reflecting a broad global public-equity owner.

That layer answers the alpha question: could a disclosure-following investor have done better than simply owning a relevant passive equity benchmark?

A later expansion adds two secondary benchmark families to ask a different question: whether any surviving strategy looks useful as a lower-risk or more institutional-style allocator even if it does not beat the market.

- Cash hurdle: BIL, used as an investable proxy for very short Treasury-bill exposure [8].
- Defensive U.S. equity comparator: USMV, a minimum-volatility U.S. equity ETF [9].
- Defensive global equity comparator: ACWV, a minimum-volatility global equity ETF [10].

These secondary comparators do not replace SPY or VT. Instead, they sit beside them. The first benchmark layer asks "did this beat the market?" The second asks "did this at least clear cash and look attractive versus a lower-volatility equity alternative?"

C. Price Sources

Price infrastructure was fund-specific:

- PIF: Twelve Data daily price history, with both raw execution prices and an adjusted mark-to-market layer [7].
- NBIM: Twelve Data adjusted daily history for the benchmark, ETF proxies, and the bounded direct mirror universe; month-end marks are derived from this daily source [7].

This unified daily source removed the earlier timing distortion from waiting until the next month-end to enter an NBIM trade and also eliminated the earlier ETF proxy gaps.

V. Strategy Definitions

All strategies in the paper are long-only and only react when a new public disclosure becomes tradable under the lag rules defined earlier. Between those dates, the portfolio is simply marked to market. No strategy assumes intraperiod foresight. The main differences between strategies are therefore:

- what information is acted on: new names, all current names, sector weights, sector changes, or cross-fund agreement;
- how capital is allocated: equal weight, disclosed weight, concentrated top sectors, or cash-retaining exposure throttles; and
- what happens after sales: immediately reinvest, rotate into other sectors, or keep the proceeds in cash.

This section is written deliberately in operational terms so that a reader can understand what each backtest is actually doing behind the shorthand labels.

A. PIF Strategies

PIF disclosures come from the reportable U.S. 13F sleeve. That means every PIF strategy begins from the same basic information set: which U.S.-listed securities were disclosed in a filing, whether they were new, continued, reduced, or exited relative to the previous filing, and when that information became public.

An important distinction is that the PIF strategies were designed for different research purposes. They are not all literal copycat portfolios. In particular:

- P1 through P4 are fully invested research variants. They do sell names when those names disappear or become disfavored, but in most cases they immediately recycle the proceeds into other eligible holdings rather than leaving the proceeds in cash.
- P5 is the cash-aware copycat variant. It is the only PIF strategy that treats disclosed selling as a reason for portfolio cash to rise rather than as a reason to concentrate more heavily in the remaining names.

This distinction matters because PIF's public sleeve often shrinks dramatically. A strategy that always reinvests sale proceeds is really testing a stock-selection rule under constant exposure, whereas a strategy that allows cash to build is testing whether the public disclosures contain useful information about overall risk posture.

Five PIF strategies were tested:

1) P1 New Positions Mirror. On each new filing date, buy only the names that newly appear in the disclosed 13F sleeve and ignore names that were already present in the prior filing. Capital is split equally across those new entries and held until the next filing-driven rebalance. At the next filing, the prior basket is sold and replaced with the new-entry basket. In plain terms, this is the purest "follow only the newly revealed ideas" test.
2) P2 Full Sleeve Equal Weight. On each filing date, hold the entire disclosed common-equity sleeve, but

give every disclosed stock the same portfolio weight regardless of PIF's position size. If a name disappears from the sleeve, it is sold; if a new name appears, it is bought; and after those changes the whole sleeve is rebalanced back to equal weights. This asks whether the mere identity of PIF's disclosed U.S. holdings has value even if the sizing information is ignored.

3) P3 Accumulation Tilt. Start from the disclosed sleeve, but overweight names that look like additions or enlargements relative to the previous filing. New positions and likely share-count increases receive more weight than unchanged positions, while exited names are removed and the capital is redistributed among the remaining eligible names. This tests whether "what PIF seems to be leaning into" is more informative than the full sleeve.
4) P4 Exit Avoidance. Start from the disclosed sleeve, but exclude or down-weight names that appear to be reductions or exits. In practice this means selling or avoiding the names that seem to be on the way out, then reallocating the capital into the names that remain eligible. This is the opposite of P3: rather than chasing additions, it attempts to avoid names that PIF seems to be backing away from.
5) P5 Cash-Aware Copy. Buy and sell the disclosed positions as they become public, but do not force sale proceeds back into the surviving names. If PIF's public sleeve shrinks, the copied portfolio can accumulate cash instead of becoming artificially more concentrated. This is the most literal "copy what was disclosed, and if they sold a lot, let cash build" interpretation.

P5 was introduced after an important conceptual observation: if PIF liquidates positions from its disclosed sleeve, a literal copycat should often keep the residual cash rather than mechanically concentrating the survivors. That distinction is important because a fully invested backtest can unintentionally turn "PIF sold several names" into "we doubled down on the names that remained," which is not the same economic signal. Put differently, P1 through P4 mostly ask "which disclosed names should we own if we insist on staying invested?", whereas P5 asks "what happens if we actually copy the observable buys and sells and allow exposure to shrink?"

B. NBIM Strategies

NBIM disclosures are fundamentally different. The dataset is much broader geographically, but it does not provide share-count history in the same way as the PIF 13F sleeve. That makes single-name mirroring much less direct and pushes the analysis toward sector and industry posture. NBIM strategies therefore fall into two families.

1) Exploratory direct mirrors:

1) N1 Core U.S. Mirror Equal Weight. Restrict NBIM to a bounded set of repeatedly observable U.S.-listed names with available price history, then hold those names equally weighted after each public release. This is a simplified stock-level copy test, but only on the observable U.S. subset rather than NBIM's full global book.
2) N2 Core U.S. Mirror NBIM Weight. Use the same bounded U.S. mirror sleeve as N1, but weight the positions in proportion to their disclosed NBIM market-value weights instead of equal-weighting them. This asks whether NBIM's stock sizing within the observable U.S. subset adds information beyond the stock list itself.

These were included because single-name mirroring was one of the original hypotheses, but they rely on a bounded U.S. mirror sleeve rather than a full global book. They are therefore useful as exploratory tests, not as the paper's cleanest evidence of investable alpha.

2) Industry-based strategies:

1) N3 Industry Weight Mirror. Translate NBIM's disclosed industry weights into liquid sector ETF proxies and hold those proxies in the same approximate proportions. This is the broadest "mirror the fund's sector posture" version.
2) N4 Industry Weight-Change Tilt. Ignore the full weight map and hold only sectors whose disclosed weights increased versus the previous snapshot. The strategy therefore acts on change, not level: it buys what NBIM appears to be increasing and ignores what is flat or shrinking.
3) N5 Industry Accumulation Tilt. Use the transition engine to build a sector score from entry/exit and accumulation/reduction counts, then hold sectors with positive net accumulation signals. This is more event-driven than N4 and leans on the transition classification rather than only the disclosed weight change.
4) N6 Top-3 Industry Leaders. At each disclosure date, hold only the three largest industry exposures in the NBIM book, implemented through sector ETFs. This is a concentrated "follow the leaders" strategy rather than a full mirror.
5) N7 Top-3 Industry Increases. Hold only the three sectors with the largest positive weight changes since the previous snapshot. This is a narrower version of N4 that focuses on the strongest rotations rather than every positive change.
6) N8 Consensus Rotation Tilt. Hold only sectors that pass both of two filters: positive disclosed weight change and positive transition-score evidence. This is designed as a higher-confidence but sparser version of the sector-rotation idea.

This second family is more realistic and more aligned with what NBIM actually discloses: slow-moving sector posture rather than actionable stock-level trading instructions.

C. Combined PIF + NBIM Strategies

The combined strategies treat the two funds as complementary signal sources rather than interchangeable portfolios. PIF contributes a risk-posture or exposure-throttle signal: should the model be fully invested, partly invested, or more defensive? NBIM contributes a sector-posture signal: if capital is deployed, which industries or sector proxies should be emphasized?

Five combined strategies were tested:

1) S1 Exposure Regime Overlay. Start with a broad global equity base, then scale gross exposure up or down using the PIF risk regime and tilt the invested sleeve toward NBIM's strongest sector signal. In plain English, this is "use PIF to decide how much risk to take and NBIM to decide where inside equities to lean."
2) S2 Cross-Fund Consensus Sector Tilt. Invest only when both funds point toward the same sector direction, and otherwise hold more cash or fallback exposure. This is the most selective confirmation strategy: it prefers fewer trades, but only when the funds line up.
3) S3 PIF Cash-Aware Base + NBIM Overlay. Start from the validated P5 cash-aware PIF portfolio and then overlay an NBIM sector tilt on top of the invested portion. This asks whether the best PIF implementation can be improved by adding NBIM's slow-moving sector view.
4) S4 N4 Sleeve With PIF Risk Filter. Start from the N4 sector-rotation sleeve and use the PIF regime only as a throttle on total exposure. The stock/sector choice comes from NBIM; PIF only decides whether the model should run at full size or partially de-risk.
5) S5 N6 Sleeve With PIF Risk Filter. Start from the concentrated N6 leadership sleeve and again use PIF only as the exposure filter. This is the concentrated leadership version of S4.

These combined strategies are important because they move the project away from the naive question "can we copy the fund?" and toward the more realistic question "can one fund help with exposure timing while another helps with sector selection?"

D. Relative Performance Conventions

The project uses two related benchmark-relative conventions, and it is important to keep them separate.

The first is a relative-NAV gap, defined as

$$R_{s,T}^{\text{relative}} = \frac{\text{NAV}_{s,T}/\text{NAV}_{s,0}}{\text{NAV}_{b,T}/\text{NAV}_{b,0}} - 1, \qquad (1)$$

where s is the strategy and b is the benchmark. This is the convention used in the core PIF and NBIM summary tables. It answers the question: after both series are rebased to 1.0, how much benchmark-relative wealth did the strategy end with?

The second is an arithmetic return gap, defined as

$$G_{s,T}^{\text{return}} = \left(\frac{\text{NAV}_{s,T}}{\text{NAV}_{s,0}} - 1\right) - \left(\frac{\text{NAV}_{b,T}}{\text{NAV}_{b,0}} - 1\right). \qquad (2)$$

This is the convention used in the robustness, attribution, and institutional-benchmark layers. It answers the question: how many percentage points of total return separated the strategy from the benchmark over the same matched window?

In plain language, the first measure is a relative-wealth comparison and the second is a return-gap comparison. Both are useful, but they should not be read as numerically interchangeable. A strategy can trail the benchmark by 25.9% on a relative-NAV basis yet be behind by 77.8 percentage points on an arithmetic total-return basis if the benchmark itself compounded much more strongly over the sample.

VI. Validation Protocol

A. Why Validation Changed the Interpretation

The project produced two major validation lessons.

1) PIF split handling: Early PIF results appeared dramatically positive. Investigation showed that raw close series were being marked against unchanged share counts through reverse splits, creating fake gains. Once the backtests were rebuilt on adjusted mark-to-market pricing, the apparent alpha disappeared and all fully invested PIF strategies became negative in benchmark-relative terms.

2) NBIM mirror-universe bias: The initial NBIM direct mirror results were also strikingly positive, but the issue was different. Here the arithmetic was fine; the interpretation was not. The bounded mirror universe turned out to be almost the same recurring basket of U.S. mega-cap winners in every snapshot. A mirror strategy built on that sleeve is closer to a hand-selected winners basket than to a neutral test of disclosure alpha.

B. Mechanical Integrity Checks

The final backtests were checked for:

- negative cash breaches,
- impossible gross exposure,
- weight-integrity gaps, and
- rebalance arithmetic consistency.

For the final NBIM panel, all strategies showed zero negative-cash breaches and only floating-point-level weight gaps. The corrected PIF engines passed the same integrity standard. A later full-project code-path audit also rechecked the original processing and baseline backtests from raw inputs forward. That audit found and fixed two genuine implementation issues in the original PIF layer: same-day bundled filings needed to be reduced to the latest observable report period before building the fully invested sleeves, and the cash-aware engine needed to prune near-zero floating-point fills so that zero-share ghost positions were not carried forward.

In addition, cached market-data rows were spot-checked against live API responses at representative benchmark, ETF, and single-name points after the final rebuild. The sampled VT, XLK, XLC, TSLA, UBER, and LCID closes matched the stored price layers exactly. A dedicated Phase-3 robustness audit then re-checked all A1–A3 outputs through 26 automated tests spanning baseline reproduction, delayed execution pricing, self-financing cost application, cap compliance at rebalance, and benchmark alignment; the final audit passed with zero failed checks.

VII. Results: PIF

A. Summary Results

Table III reports the validated PIF strategy outcomes. The key conclusion is more nuanced than the earlier draft of this project suggested. Once the original bundle-handling bug in the fully invested strategies is corrected, P2 through P5 all generate positive absolute returns. However, none of them produce alpha versus SPY, so the public PIF sleeve still fails as a reliable benchmark-beating copy-trading signal.

B. Interpretation

Three findings are robust:

1) Absolute return and alpha are different here. P2 through P5 all make money in dollar terms over the matched sample, but none of them beat passive U.S. beta. The strongest absolute PIF result, P2, still trails SPY by about 25.9% on a relative-NAV basis.
2) Cash-awareness is economically informative, but not dominant. P5 remains the most literal copycat design because it allows cash to build when the disclosed sleeve shrinks. That makes it useful for interpreting PIF's public filings as an exposure signal, even though it is not the highest-return PIF strategy in the sample and still does not generate alpha.
3) Partial disclosure is a structural limit. Even a mechanically faithful mirror of PIF's 13F book is not a mirror of the entire fund.

The implication is that PIF's public sleeve is better interpreted as a monitoring and posture dataset than as a direct alpha source. The fully invested variants show that the disclosed sleeve can still participate in broad equity upside, but the lagged public information is not strong enough to beat a passive benchmark once the sleeve is implemented honestly.

VIII. Results: NBIM

A. Exploratory Mirrors versus Realistic Signals

Table IV reports the final NBIM strategy results versus VT. Unlike the PIF panel, the table must be interpreted in two layers: the exploratory direct mirrors and the realistic sector-based tests.

B. Why N1 and N2 Are Not Strong Alpha Evidence

The most important NBIM result is not the high return of N1 and N2, but the reason that those returns cannot be treated as robust disclosure alpha. The mirror universe was a bounded U.S. mega-cap basket, and the bias audit showed that it overlapped almost perfectly with the top recurring U.S. names in the NBIM book across the sample. In other words, the backtest was effectively long a persistent basket of secular winners.

That is a valid exploratory result—it tells us what a bounded, disclosure-informed mega-cap sleeve would have done—but it is not clean evidence that the public disclosure stream itself creates a broadly harvestable copy-trading edge.

C. What Looks More Credible

The realistic NBIM findings are much narrower:

- N3 is essentially benchmark-like. Broadly mirroring NBIM's sector posture mostly reproduces a high-quality global allocation.
- N4 is the strongest realistic signal. Following sectors whose disclosed weights increased produced a modest positive relative gap of 18.6% versus VT in the core NBIM summary layer.
- N6 also worked reasonably well. Concentrating on NBIM's top three sector exposures produced an 18.1% excess return.
- N5, N7, and N8 are weak or too sparse. Transition-only and consensus-only rules do not appear strong enough on their own in this implementation.

Thus, the most credible NBIM signal is not direct mirroring but slow-moving sector posture, especially disclosed industry weight change.

IX. Results: Combined PIF + NBIM Strategies

A. Phase-2 Combined Results

The Phase-2 tests ask a narrower and more realistic question than either single-fund panel alone: can PIF's exposure-regime information and NBIM's sector-posture information be combined into a modest but credible overlay strategy? Table V summarizes the results.

Three observations stand out.

1) S1 is the only positive VT-relative result. The combination of a PIF risk regime and an NBIM sector overlay can add a small amount of value versus VT, but the edge is modest.
2) Consensus-only confirmation is too restrictive. S2 spends too much time in cash or fallback exposure, which limits both upside and signal harvest.
3) PIF helps more as a throttle than as a selection engine. S4 and S5 are cleaner than S3 because they preserve validated NBIM sleeves and only use PIF to modulate gross exposure. Even so, they still fail to beat VT.

The combined evidence therefore strengthens, rather than weakens, the conservative interpretation of the project. There is some informational value in the interaction between the two funds, but it appears to be a risk-and-sector overlay effect rather than a strong source of standalone alpha.

B. Phase-3 Robustness Results

The next question is whether any surviving signals remain after more realistic implementation stress. Four focus strategies were selected for robustness work:

TABLE III
PIF Strategy Results versus SPY

Strategy	Total Return	CAGR	Max Drawdown	Excess Return vs SPY	Information Ratio
P1 New Positions Mirror	−75.8%	−20.2%	−94.8%	−90.5%	−0.299
P2 Full Sleeve Equal Weight	+122.3%	+11.6%	−62.2%	−25.9%	+0.015
P3 Accumulation Tilt	+68.6%	+7.5%	−65.4%	−43.8%	−0.084
P4 Exit Avoidance	+78.1%	+8.3%	−65.4%	−40.6%	−0.064
P5 Cash-Aware Copy	+48.9%	+5.6%	−62.6%	−50.4%	−0.120

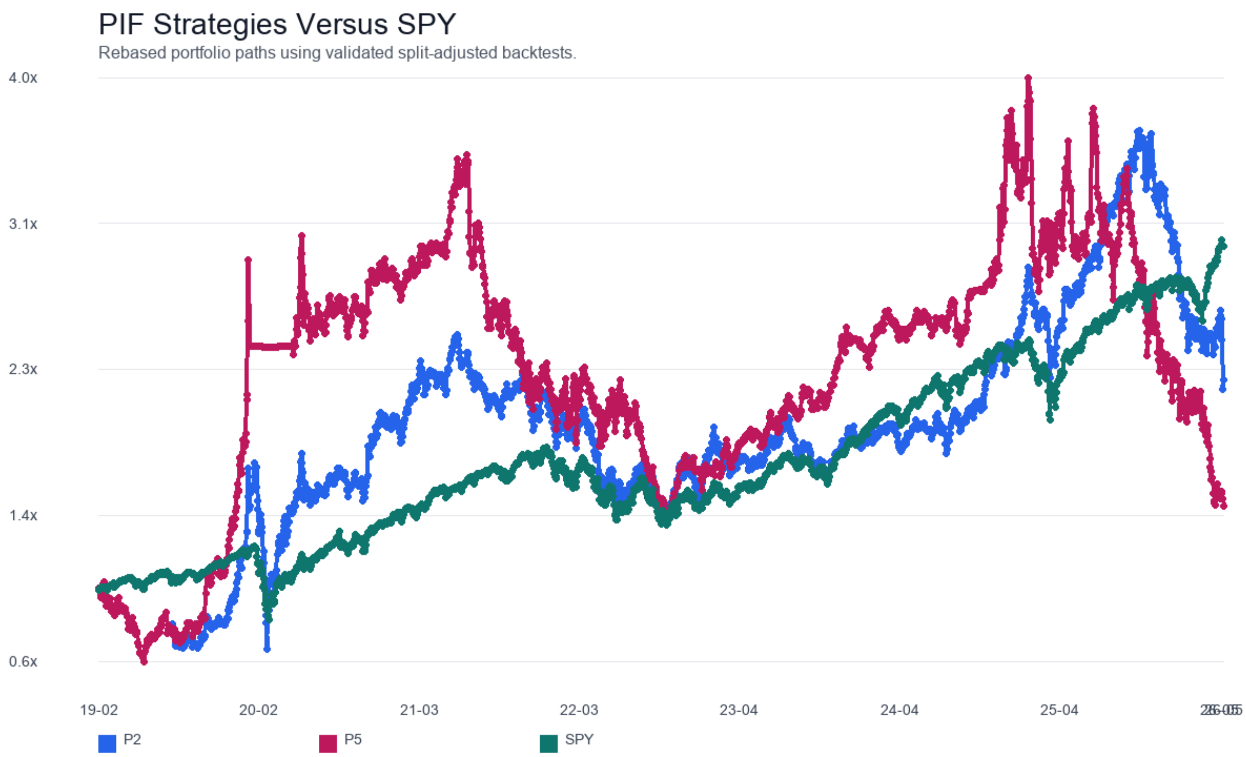

Fig. 1. Validated PIF strategy paths versus SPY. After correcting same-day bundle handling in the fully invested strategies, P2 is the strongest absolute-return PIF variant. Even so, every PIF strategy still lags SPY on a matched excess-return basis.

TABLE IV
NBIM Strategy Results versus VT

Strategy	Type	Total Return	Ann. Return	Max Drawdown	Excess Return vs VT
N1 Core U.S. Mirror Equal Weight	Exploratory mirror	+717.0%	33.6%	−28.6%	+231.7%
N2 Core U.S. Mirror NBIM Weight	Exploratory mirror	+466.8%	27.0%	−34.7%	+119.8%
N3 Industry Weight Mirror	Broad industry mirror	+161.1%	14.2%	−22.2%	+2.1%
N4 Industry Weight-Change Tilt	Targeted rotation	+199.0%	16.3%	−24.3%	+18.6%
N5 Industry Accumulation Tilt	Targeted transition tilt	+69.2%	7.5%	−24.3%	−32.9%
N6 Top-3 Industry Leaders	Targeted concentration	+197.7%	16.2%	−25.8%	+18.1%
N7 Top-3 Industry Increases	Targeted rotation	+129.9%	12.2%	−24.4%	−8.8%
N8 Consensus Rotation Tilt	Targeted consensus filter	+83.3%	8.7%	−19.2%	−27.3%

- P5: the best PIF design conceptually, because it retains cash after disclosed sales;
- N4: the strongest NBIM weight-change strategy;
- N6: the strongest concentrated NBIM leadership sleeve; and
- S1: the best combined PIF + NBIM overlay.

Three robustness dimensions were tested:

1) A1 execution lag: baseline next-day trading versus T+3 and T+5;
2) A2 explicit one-way execution costs: 0, 10, 25, and 50

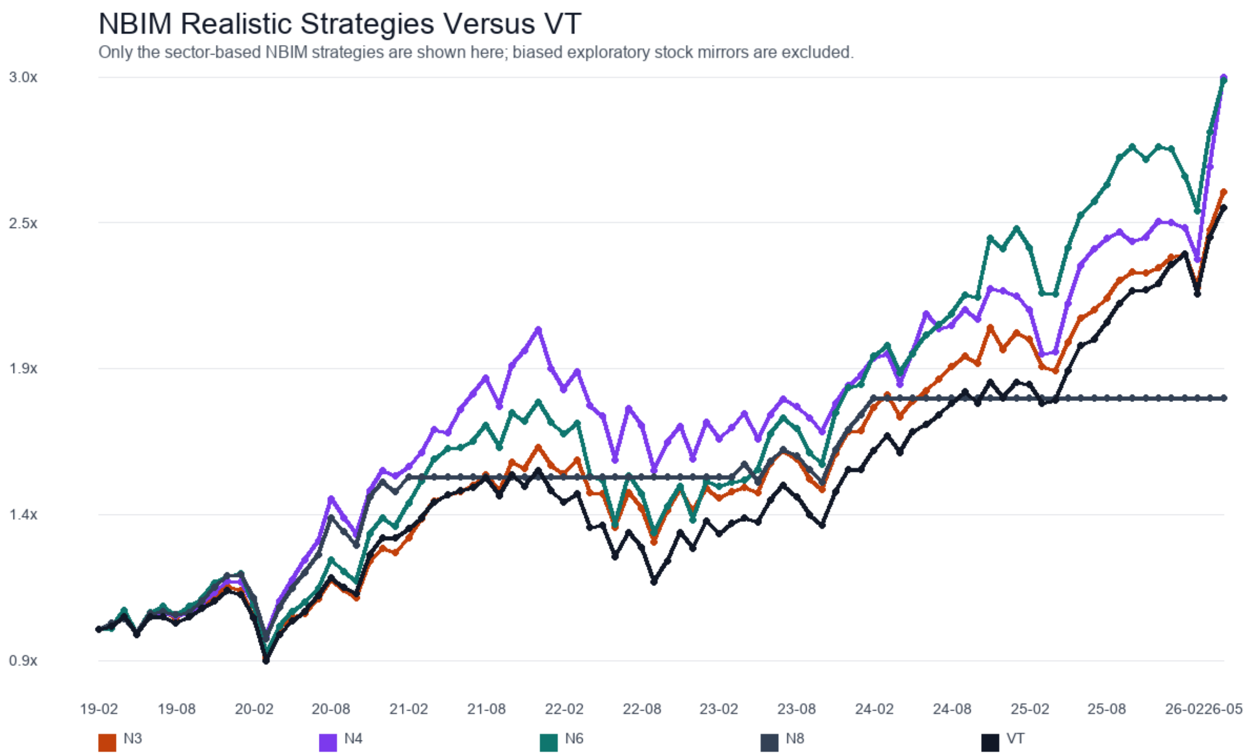

Fig. 2. NBIM sector-based strategies rebased against VT. Only the realistic industry strategies are shown; the exploratory direct mirror sleeve is intentionally excluded from this figure.

TABLE V
Combined PIF + NBIM Strategy Results

Strategy	Total Return	Excess Return vs VT	Excess Return vs SPY	Max Drawdown	Avg. Cash Weight
S1 Exposure Regime Overlay	+154.7%	+1.9%	−15.1%	−25.1%	27.0%
S2 Cross-Fund Consensus Sector Tilt	+82.3%	−27.1%	−39.2%	−22.1%	42.4%
S3 PIF Cash-Aware Base + NBIM Overlay	+58.0%	−36.8%	−47.3%	−60.2%	7.8%
S4 N4 Sleeve With PIF Risk Filter	+134.4%	−6.2%	−21.9%	−24.3%	27.6%
S5 N6 Sleeve With PIF Risk Filter	+133.9%	−6.4%	−22.0%	−29.4%	27.2%

bps;

3) A3 concentration and exposure caps: realistic position, sector, and gross-exposure limits.

Table VI compresses the most decision-relevant results. It reports the baseline benchmark-relative excess return together with the same measure under the most demanding lag, cost, and cap variants tested.

The interpretation is sharper than the Phase-2 results alone:

- P5 remains non-alpha. The strategy is still useful for understanding PIF as an exposure-throttle signal, but it does not beat SPY under any tested implementation variant.
- N4 is plausible but cap-sensitive. Its positive excess return survives delayed execution and one-way costs, but disappears once realistic sector caps are imposed.
- N6 is the strongest residual signal. It remains positive versus VT under delayed execution, explicit costs, and both moderate and tight cap regimes.
- S1 is fragile. The modest Phase-2 edge vanishes once execution is delayed or even modest costs and exposure constraints are imposed.

This robustness layer matters because it narrows the project's actionable conclusion. The surviving signal is not "copy SWFs" in general, nor even "combine PIF and NBIM." It is specifically that a simple, slow-moving NBIM sector-leadership sleeve appears to carry some residual information even after several realistic implementation stresses.

The subsequent subperiod-stability test reinforces the same hierarchy. P5 is strong only in the early sample and gives back the edge in 2024–2026. S1 is positive in 2019–2021 but negative in the later subperiods. N4 is regime-dependent: strong in 2019–2021, negative in 2022–2023,

TABLE VI
Phase-3 Robustness Summary for the Focus Set

Strategy	Benchmark	Baseline Excess	T+5 Excess	50 bps Excess	Tight-Cap Excess
P5 Cash-Aware Copy	SPY	−151.1%	−149.0%	−161.2%	−156.7%
N4 Industry Weight-Change Tilt	VT	+71.2%	+70.8%	+52.1%	−48.8%
N6 Top-3 Industry Leaders	VT	+64.4%	+62.9%	+60.0%	+33.7%
S1 Exposure Regime Overlay	VT	+4.7%	−8.7%	−14.3%	−79.0%

and only modestly positive thereafter. N6 is the only focus-set strategy that remains positive versus VT in all three subperiods, although its excess return becomes very small in the most recent window. This is not evidence of a large persistent alpha engine, but it is stronger support for N6 than for any other tested strategy in the project.

C. Benchmark Robustness and Attribution

Phase 3 then asked whether the remaining conclusions depend too heavily on benchmark choice. This matters especially for two cases: P5, where a U.S. 13F sleeve might look different against growth-heavy benchmarks, and S1, where a mixed cash-and-sector strategy might appear stronger against a global benchmark than against a U.S. benchmark.

The benchmark-robustness results sharpen the interpretation further:

- P5 remains non-alpha under both benchmark choices. It underperforms SPY by 151.1% and underperforms QQQ by 282.9% on the matched live window.
- N4 and N6 remain positive under an alternate global benchmark. On the arithmetic return-gap convention used in the robustness layer, N4 posts 71.2% versus VT and 69.4% versus ACWI. N6 posts 64.4% versus VT and 62.7% versus ACWI.
- S1 is benchmark-sensitive. It is slightly positive versus VT at 4.7%, but negative versus SPY at −45.4% and still negative versus a 50/50 rebased VT/SPY blend at −20.4%.

That result is important because it separates “global-allocation-like” behavior from true cross-market alpha. S1 appears closest to a modest global-overlay effect, not to a benchmark-agnostic source of excess return.

D. Institutional Benchmarking and Consistency Metrics

The opportunity-cost benchmark remains the right test for alpha, but many real allocators are judged on a broader standard: do they produce acceptable returns while preserving capital more effectively than a high-beta equity sleeve? To test that question directly, the project adds a three-layer benchmark stack for the strongest representative strategies:

- P2, because the original-engine audit showed it is the strongest absolute-return PIF sleeve;
- P5, because it is the most literal cash-aware PIF interpretation;
- N4 and N6, because they are the strongest realistic NBIM sleeves; and
- S1, because it is the most interpretable cross-fund overlay.

For each strategy, the paper then measures performance against:

1) the original market benchmark;
2) the BIL cash hurdle; and
3) the relevant defensive-equity comparator (USMV for PIF, ACWV for NBIM and combined strategies).

In addition to total return and excess return, the layer computes Calmar ratio, Sortino ratio versus the cash hurdle, monthly downside and upside capture versus the market benchmark, positive-month rate, worst rolling loss windows, and recovery duration.

In this table, “return gap” means the arithmetic difference in matched total returns, not the relative-NAV measure used in the earlier fund-level summary tables. “Downside capture” below one means the strategy lost less than the market benchmark in down months, while values above one mean it lost more.

This expanded lens changes the interpretation in a useful way:

- PIF mirroring still does not look institutionally compelling. P2 does beat both cash and USMV in absolute return, but it does so with annualized volatility above 35%, a drawdown above 62%, and downside capture above one. That is not the profile of a smoother institutional allocator. P5 lowers downside capture dramatically, but still trails USMV and retains a similarly deep drawdown.
- N4 and N6 both remain strong against cash and defensive equity comparators. N4 has the best cash-relative Sortino ratio and downside capture below one, while N6 remains the stronger return-seeking sleeve with a higher positive-month rate and stronger upside participation.
- S1 looks better as a lower-beta overlay than as an alpha engine. Its arithmetic market return gap versus VT is only modest, but it remains strongly positive versus both BIL and ACWV, with downside capture around 0.74 and upside capture around 0.85. In plain terms, S1 looks more like a controlled participation overlay than like a high-conviction benchmark-beating portfolio.

TABLE VII
Expanded Benchmark and Consistency Readout

Strategy	Market Bench	Return Gap vs Market	Return Gap vs Cash	Return Gap vs Defensive	Sortino vs Cash	Downside Capture	Max Drawdown
P2 Full Sleeve Equal Weight	SPY	−77.8%	+101.8%	+29.5%	0.59	1.12	−62.2%
P5 Cash-Aware Copy	SPY	−151.1%	+28.5%	−43.8%	0.39	0.23	−62.6%
N4 Industry Weight-Change Tilt	VT	+71.2%	+199.2%	+147.7%	1.34	0.88	−24.9%
N6 Top-3 Industry Leaders	VT	+64.4%	+192.4%	+140.9%	1.11	1.08	−24.3%
S1 Exposure Regime Overlay	VT	+4.7%	+134.2%	+89.6%	0.95	0.74	−25.1%

This benchmark expansion is important because it prevents a false binary conclusion. A strategy can fail as a market-alpha engine yet still be informative as a lower-beta overlay. In this project, however, the evidence remains selective even under that friendlier lens: PIF mirroring is still weak, N4 and N6 are still the best-returning realistic sleeves, and S1 is best understood as a cautious cross-fund overlay rather than as a flagship alpha model.

E. Formal Statistical Inference and Turnover-Adjusted Metrics

The robustness tests above are still mainly empirical. A natural next question is whether the surviving effects are statistically convincing once one tests the realized excess-return series directly rather than only inspecting terminal backtest outcomes.

To answer that, the project adds a final inference layer on top of the validated strategy-versus-benchmark timelines. For each focus-set strategy, period excess returns are reconstructed directly from the already-audited benchmark-comparison files. Three diagnostics are then applied:

1) a Newey–West mean-excess test on the realized excess-return series;
2) a moving-block bootstrap confidence interval for annualized excess return; and
3) a block sign-permutation test as a non-parametric robustness check.

In parallel, the already-validated 10 bps cost layer is used to compute turnover-adjusted Sharpe and information ratios rather than relying only on total-return drag.

This table changes the tone of the project in an important way:

- No focus-set strategy clears a strong conventional significance threshold. Even N6, the strongest residual survivor, has a Newey–West p-value around 0.12 and a bootstrap interval for annualized excess return that still crosses zero.
- N6 remains the cleanest implementation candidate anyway. It has the best turnover-adjusted information ratio and by far the lowest annualized trading burden among the surviving positive strategies.
- N4 is economically interesting but statistically weaker than its raw excess total suggests. This is consistent with the earlier concentration and cap-sensitivity results.
- S1 is not just fragile economically; it also lacks inferential support. Its annualized excess mean is essentially zero, and its turnover-adjusted information ratio turns slightly negative at the 10 bps implementation level.

In other words, the inference layer does not overturn the economic ranking of the strategies, but it does downgrade the strength of the claim one can make. The best remaining result is better described as evidence of a modest, plausible residual effect than as proof of a durable alpha engine.

The attribution layer then explains why the four focus strategies behave differently. The decomposition is performed in exact daily arithmetic excess-return space, where each period's excess return is split into:

1) an exposure timing effect, driven by holding more or less benchmark exposure through cash;
2) an allocation effect, measured by equal-weighting the invested sleeve; and
3) a concentration effect, measured as the incremental impact of the actual strategy weights versus equal weights within that same sleeve.

The resulting interpretation is highly consistent with the earlier robustness findings:

- P5 fails mostly because of concentration, not because of cash alone. Its cumulative arithmetic cash drag is small at roughly −0.9 percentage points, while the concentration term is strongly negative at about −105.7 percentage points.
- N4 relies heavily on concentration. Its allocation term is negative, but a strongly positive concentration term overwhelms it. This fits the cap-sensitivity finding from A3.
- N6 is the cleanest survivor. Its allocation term is strongly positive, while concentration is near zero. That is exactly the pattern one would want from a robust sector-leadership sleeve.
- S1 gives back much of its potential edge through cash drag. It has a positive allocation term, but a large negative exposure-timing effect from running below full benchmark exposure.

This attribution result is arguably the most decision-useful output of Phase 3. It shows that N6 is not merely surviving by accident or by leverage-like concentration. Instead, its residual signal appears to come from the sectors it chooses to own rather than from aggressive weight concentration.

TABLE VIII
Formal Inference and Turnover-Adjusted Metrics for the Focus Set

Strategy	Benchmark	Ann. Excess Mean	NW p-value	95% Bootstrap CI	10 bps IR	Ann. Gross Turnover
P5 Cash-Aware Copy	SPY	−4.3%	0.745	[−31.3%, 22.2%]	−0.125	4.70×
N4 Industry Weight-Change Tilt	VT	+3.3%	0.292	[−2.7%, 9.8%]	+0.331	2.86×
N6 Top-3 Industry Leaders	VT	+3.4%	0.122	[−1.1%, 7.6%]	+0.498	0.45×
S1 Exposure Regime Overlay	VT	+0.1%	0.984	[−6.2%, 6.4%]	−0.017	3.54×

F. Event-Window Evidence

The final Phase 3 diagnostic asked a narrower question: which individual state changes are actually followed by favorable forward returns? This event-window layer was built directly from the validated signal-state tables rather than from the portfolio backtests themselves. Six event families were tested over one-, three-, and six-month windows:

- transitions into PIF expanding;
- transitions into PIF contracting;
- NBIM technology overweight events;
- all positive NBIM industry-weight-change events;
- cross-fund consensus gained events; and
- cross-fund consensus lost events.

The audit for this step is clean. The event-window output contains 236 realized windows and no failed logic checks; the only non-pass audit rows are four explicit end-of-sample skips for the final 2026 technology event where the full three- and six-month horizons are not yet observable. The event table was then hardened one step further by attaching unconditional baseline returns for the same proxy asset and horizon. This makes it possible to compare state-conditioned forward returns not only to cash or VT, but also to the ordinary drift of the same underlying asset over the broader sample.

The results are informative precisely because they are not overly flattering:

- PIF regime changes are not strong directional timing signals on their own. Both PIF expanding and PIF contracting events are followed by positive absolute SPY returns on average. The six-month average forward return is about 13.7% after expanding events and 11.9% after contracting events. When compared with unconditional SPY drift over the same six-month horizon, both are still above average, but expanding remains only modestly stronger than contracting. That weakens any interpretation that the PIF state machine alone offers a clean directional market-timing edge.
- NBIM technology overweight events are directionally consistent with the stronger N6 result. The technology overweight event family produces a positive six-month average excess return of about 6.5% versus VT, with a 71.4% excess hit rate.
- All positive NBIM industry-weight changes are only mildly useful in aggregate. Across all sectors, the six-month average excess return is roughly 1.4%, and the three-month average excess return is slightly negative. This suggests that positive weight changes are not uniformly informative and need sector selection.
- The sector breakdown matters. Within the positive-industry-change family, Energy and Technology are the strongest six-month sectors, while Financials and Communication Services are weak in this sample. That again points toward selective leadership effects rather than broad portfolio-wide timing skill.
- Cross-fund consensus is better as confirmation than as a reversal signal. Consensus gained events are modestly positive at six months, with average excess return near 4.9% versus VT. But consensus lost events are not reliably negative; they remain slightly positive on average. In other words, gaining alignment may help identify constructive sectors, while losing alignment is not a clean short or de-risking trigger.

Once bootstrap confidence intervals are added to these event families, the interpretation becomes even more disciplined. At the six-month horizon, every family-level excess above unconditional baseline interval still overlaps zero, including PIF expanding, PIF contracting, NBIM overweight tech, and cross-fund consensus gained. That does not mean the events are useless; it means the evidence is better treated as directional and suggestive than as a statistically settled timing rule. The event-window layer therefore supports the same high-level conclusion as the strategy inference layer: the surviving effects are real enough to keep studying, but not strong enough to justify overconfident alpha claims.

This event-window layer reinforces the broader project conclusion. The residual signal is not a universal “follow every state change” rule. It is narrower and more selective: technology-heavy or leadership-style NBIM sector signals survive best, while the broader PIF regime shifts and consensus-lost conditions are too blunt to support strong standalone timing claims.

G. Window Hit Rates and Contribution Concentration

The next diagnostic asks whether the surviving strategies work broadly across many rebalance windows or whether their final results are dominated by a handful of periods. This is a useful complement to the event-window analysis because it stays at the portfolio level while still decomposing performance into discrete, auditable windows.

For each focus-set strategy, the analysis partitions the validated benchmark-comparison timeline into rebalance-to-rebalance windows. For every window it records:

- the strategy window return;
- the benchmark window return;
- the excess window return; and
- the exact additive change in relative excess NAV over that window.

That final quantity is especially important because it sums exactly to the terminal benchmark-relative NAV difference. This makes concentration analysis truthful rather than approximate.

The resulting pattern again supports the same hierarchy:

- P5 is highly concentrated and unstable. It beats SPY in only 39.1% of rebalance windows. Its top three positive windows account for roughly 78.6% of total positive contribution, and the single largest positive window alone accounts for more than 41%. This is not the profile of a broad or repeatable alpha process.
- N4 remains positive, but the contribution profile is concentrated. It beats VT in 54.5% of windows, yet its top three positive windows account for about 82.5% of total positive contribution. This is consistent with the earlier interpretation that N4 works partly through concentrated winner episodes.
- N6 is the broadest surviving effect. It beats VT in 63.6% of windows, and its positive-contribution concentration is lower than N4's. The top three positive windows still matter, but they account for a smaller share of the total, and the positive-contribution HHI is meaningfully lower. This is further evidence that N6 is not merely a lucky concentration outcome.
- S1 is diffuse but weak. Its contribution concentration is relatively low, which means the strategy is not driven by only a few spikes. But its benchmark-beating hit rate is just 43.2%, and the final relative excess remains small. This fits the earlier attribution result: there is some allocation value in S1, but it is diluted by underinvestment and cash drag.

This is a useful distinction. A strategy can be broad-based but weak, or strong but highly concentrated. N6 is the only focus-set strategy that looks at least somewhat favorable on both dimensions at once: it survives the robustness tests, shows a positive hit-rate profile, and is not overwhelmingly dominated by a single small cluster of windows.

H. Formal State Model and Persistence

The final Workstream in Phase 3 converts the validated cross-fund signal layer into a compact state model that is easier to monitor and reason about than the full set of backtests. This step matters because the project is no longer only about whether a specific portfolio rule earned excess return. It is also about whether the underlying disclosure system can be summarized into durable and interpretable market states.

The state table is built directly from the validated combined signal-state layer. Each state date records:

- a PIF risk state, mapped from initial, expanding, stable, and contracting into risk_on, neutral, and risk_off;
- an NBIM sector state, classified as consensus_led, overweight_led, or top3_led;
- the number of cross-fund confirming sectors;
- a model exposure target derived from the PIF state; and
- a primary sector tilt derived from the strongest cross-fund consensus sector, the strongest NBIM overweight sector, or the strongest NBIM top-three sector.

The resulting compact table contains 38 state dates and 27 explicit state changes. The audit for this layer is clean, with no failed logic checks. The structure of the state space is informative on its own:

- The model is mostly sector-led rather than broad-market-led. Nineteen of the 38 dates are classified as consensus_led, 17 as overweight_led, and only two as pure top3_led fallback states.
- Technology dominates the primary tilt. Technology is the primary modeled sector on 21 of the 38 dates, followed by Financials on nine dates, Consumer Discretionary on five, and Communication Services on three.
- Risk-on periods are shorter lived than neutral or risk-off periods. The average closed duration for a risk_on state is about 71 days, versus about 102 days for neutral and about 183 days for risk_off. In other words, the constructive phases appear more tactical, while defensive states are more persistent.

This persistence result is especially useful because it supports the interpretation developed earlier in the paper. If the states were flipping every few days, the whole framework would look too noisy to monitor. Instead, the dominant configurations persist for weeks to months, which makes them plausible research states rather than micro-timing artifacts.

I. Forward Returns by State

The next question is whether those compact states are useful directly, rather than only through full portfolio backtests. To answer that, each state date is evaluated over one-, three-, and six-month forward windows using the same validated daily benchmark and sector-proxy histories. For each state, the system records:

- the forward return of SPY;
- the forward return of SPY relative to unconditional SPY drift over the same horizon;
- the forward return of VT; and
- when a primary sector proxy exists, the forward excess return of that sector ETF versus VT, both raw and relative to the unconditional average excess of that same sector.

This layer also audits cleanly. There are no failed rows. The only non-pass records are expected skips caused either by the pre-SPY start date of the earliest 2018 state or by incomplete end-of-sample windows in 2026.

The resulting interpretation is conservative but useful:

- Risk-on states are directionally constructive, but not explosively so. At the six-month horizon, risk_on states are followed by SPY returns that are about 3.0 percentage points above unconditional six-month SPY drift.
- Risk-off states are not simple bearish market calls. Six-month SPY returns after risk_off states are also above unconditional drift, at about 3.4 percentage points. This is consistent with the earlier event-window result that PIF contracting is not a clean directional market-timing signal by itself.
- Technology-led states are the most stable constructive sector family. Six-month Technology-led states deliver positive sector excess versus VT, and they remain slightly above unconditional Technology leadership as well.
- Financials- and Consumer-Discretionary-led states are weaker. Their six-month primary-sector excess returns are negative relative to unconditional sector baselines, even when the accompanying SPY drift is still positive.
- Communication-Services-led states are unusual but informative. They are associated with weak broad-market drift but strong sector-specific outperformance relative to unconditional Communication Services behavior. This looks more like a narrow leadership condition than a market-level risk-on regime.

Taken together, the state layer clarifies the practical meaning of the broader project. The useful information is not "follow every sovereign wealth fund move" and not even "buy every positive state." Instead, the most credible use case is to treat the system as a slow-moving allocator: PIF helps define exposure posture, NBIM helps define sector posture, and the strongest surviving state family is still the technology-heavy leadership complex that also underlies the better NBIM strategy results.

X. Discussion

A. What the Combined Evidence Says

The PIF and NBIM panels point to the same high-level conclusion from different directions:

- direct single-name copy trading is weak once lag and partial visibility are respected;
- broad mirroring often collapses into benchmark-like exposure rather than differentiated alpha; and
- the most plausible residual signal lies in slower-moving allocation, sector rotation, and risk throttling.

In PIF, even the most sensible copycat variant fails to beat SPY. In NBIM, the strongest realistic sector-based strategies are positive, but only modestly so. In the combined panel, only S1 exceeds VT, and only by a small amount; the filtered-sleeve variants S4 and S5 remain below both VT and SPY.

Phase 3 further tightens that conclusion. Once lag sensitivity, explicit costs, concentration constraints, subperiod stability, and alternate benchmark choices are modeled, the combined edge in S1 no longer looks robust. N4 remains interesting but cap-sensitive and concentration-dependent. N6 is the only strategy in the focus set that stays meaningfully positive through all tested robustness layers, and its attribution profile is cleaner than the alternatives.

For that reason, the final model expression of the project is intentionally simpler than the full strategy library. The best candidate investable sleeve is still N6 itself, while the best higher-level product abstraction is the cross-fund state model. The expanded institutional benchmark layer adds one refinement: S1 is worth keeping as a lower-beta overlay reference even though it is not strong enough to be the flagship alpha sleeve. No new hybrid production strategy was added at the end of the project because the hardening layer weakened the case for more complexity rather than strengthening it.

B. What Was Built Beyond the Final Returns

This project produced more than terminal backtest numbers:

- normalized holdings and transition datasets for PIF and NBIM;
- audit layers for disclosure dates, price coverage, and mirror-universe bias;
- benchmark-relative reports and chart packs;
- a PIF interactive explorer for stepping through filings, signals, orders, and portfolio evolution, later upgraded with plain-English strategy rulebooks and filing-by-filing explanation panels; and
- a reusable structure for extending the project to future monitoring and higher-capacity pricing sources.

This matters because disclosure-based alpha is especially vulnerable to hidden assumptions. The system-level outputs were designed to make those assumptions inspectable.

XI. Assumptions and Caveats Appendix

Table IX summarizes the assumptions and scope limits that most directly shape the interpretation of the project. The point of this table is not to weaken the work, but to make the boundaries of the claims explicit and easy to audit.

XII. Limitations

Several limitations remain:

- market impact, taxes, and venue-specific implementation frictions were not modeled even though Phase 3 did add explicit lag, transaction-cost, cap, subperiod, and benchmark sensitivity;

TABLE IX
Assumptions and Caveats Summary

Category	Assumption or Caveat	Scope	Why It Matters
Disclosure timing	Trades may occur only after public release and never on the underlying portfolio effective date unless that date is also public.	PIF and NBIM	This is the core protection against look-ahead bias.
Disclosure coverage	PIF covers only the reportable U.S. 13F sleeve.	PIF	The copied book is not the whole sovereign fund, so underperformance cannot be interpreted as the full fund failing.
Disclosure coverage	The NBIM direct mirror uses a bounded observable U.S. sleeve rather than the full global book.	NBIM direct mirrors	This can create survivor-selection bias and is why the direct mirror results are treated as exploratory only.
Signal translation	NBIM sector posture is expressed through U.S. sector ETF proxies.	NBIM and combined strategies	This introduces proxy error between a global industry book and U.S.-listed sector instruments.
Benchmark proxies	Cash and defensive benchmark layers are implemented with investable ETF proxies rather than with official total-return index series.	Phase 4 benchmark extension	The comparative ranking is still useful, but the exact hurdle levels reflect ETF implementation rather than frictionless index returns.
Pricing	Portfolio valuation uses adjusted prices.	PIF and NBIM	This avoids split-driven false gains and keeps the mark-to-market layer economically coherent.
Friction model	Costs are modeled as one-way basis-point charges, but market impact and taxes are omitted.	Phase 3 focus set	The live implementation burden could still be worse than the modeled drag.
Inference	Strategy inference is run on realized excess-return series with HAC and bootstrap methods.	Phase 3 focus set	Large terminal returns do not automatically imply strong statistical evidence.
Inference	Event-window families use bootstrap confidence intervals.	Event-window layer	Small event counts can produce unstable averages, so event interpretations should remain cautious.
State model	PIF risk states are mapped into fixed exposure targets of 1.00, 0.75, and 0.50.	Combined state model	This is a monitoring abstraction, not proof that those exact sizing levels are optimal.
Portfolio rules	All strategies are long-only and unlevered.	All backtests	This keeps the tests interpretable and investable, but may understate what a more flexible allocator could do.
Scope	GIC remains qualitative only.	Overall research design	This avoids false precision from sparse public data but leaves one major SWF outside the quantitative panel.

- GIC was not included quantitatively because the public data was too incomplete;
- NBIM sector tests still used U.S. ETF proxies for a global industry book, introducing representation error;
- the NBIM direct mirror universe remained bounded and should still be treated as exploratory even after the timing and daily-price corrections;
- the strongest NBIM direct mirror results were contaminated by mirror-universe selection bias;
- the attribution layer is exact in arithmetic excess-return space, but it should not be confused with a perfect decomposition of compounded terminal return.

These do not invalidate the project, but they sharply limit how strong any alpha claim can be.

XIII. Conclusion

This study asked whether sovereign wealth fund disclosures can be monetized after realistic lag and visibility constraints are enforced. The answer is mostly negative for naive copy trading and only cautiously positive for narrow sector-based signals.

For PIF, no strategy generated alpha versus SPY. After the full audit, the absolute-return leader is the full-sleeve equal-weight design (P2), not the cash-aware design (P5). That revision matters because it changes the economic reading: cash-awareness remains useful for interpreting PIF as an exposure signal, but the strongest realized PIF result in the sample still comes from a fully invested sleeve that simply fails to beat passive beta.

For NBIM, the strongest raw stock-mirror result could not be accepted at face value because it was driven by a bounded basket of recurring mega-cap winners. More realistic industry-based strategies were far more credible, but the edge was modest. The best signals came from industry weight changes and concentrated leadership sectors, not from broad portfolio mirroring.

For the combined Phase-2 layer, the conclusion is similar. Using PIF as a gross-exposure regime and NBIM as a sector allocator can produce a small benchmark-relative edge versus VT, but the effect is not large enough to support a strong alpha claim and does not survive the tougher implementation tests in Phase 3. The expanded benchmark layer does, however, clarify what S1 is: not a strong alpha engine, but a lower-beta overlay that remains comfortably ahead of the cash hurdle and the defensive global-equity comparator while participating less than one-for-one in market drawdowns.

The highest-confidence conclusion is therefore:

> Public SWF disclosures are more useful as slow-moving allocation and rotation signals than as literal copy-trading instructions, and even then the most durable evidence in this project narrows to a modest NBIM sector-leadership effect plus a lower-beta cross-fund overlay concept, rather than a broad copy-trading alpha engine.

The practical continuation of this research should prioritize event-level attribution, state-transition analysis, and monitoring tools, rather than ever-larger naive mirroring engines.

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